

Interactive MObjects Tester: Code Process and Flow

Implementation walkthrough for maintaining and extending the new tester.

Design intent: MObjects is isolated behind a product-producing adapter so SEP, MObjects, and future detectors can be compared through common masks, rows, profiles, and PNG outputs.

Runtime Entry Point

1. `parse_args` reads `--manifest`, `--pc`, and `--mobjects-root`.
2. `main` creates `MObjectsTester` with the selected manifest, machine, and MObjects checkout path.
3. `MObjectsTester` loads the local manifest through `interactive_galclean_parameter_tester` helpers.
4. The Tkinter UI exposes machine, galaxy, detection surface, units, MObjects parameters, spike-gate controls, and post-filters.

Calculation Flow

Stage	Function	Responsibility
Load FITS	load_fits / _load_galaxy	Read image data and geometry, with a per-galaxy cache.
Detection image	prepare_detection_image	Create original or smoothed-residual detection image and non-finite mask.
MTOjects setup	mtobjects_context	Temporarily switch into the MTOjects checkout so its compiled libraries load correctly.
Parameter object	mtobjects_parameter_namespace	Build the SimpleNamespace expected by MTOjects internals.
Tree detection	mtobjects_products	Preprocess image, build max tree, filter tree, relabel segments, and produce raw rows.
Post-filtering	measure_segments / filter_segmentation	Measure labels and apply min/max area, elongation, centre exclusion, and dilation.
Spike gate	spike_gated_mtobjects_products	Run the gate MTOjects pass and keep labels intersecting detected bar-profile spikes.
Visual output	draw_products	Render original, residual, isophotes, profiles, masks, and spike samples; save a PNG.

Product Contract

Both the normal MTOBJECTS run and spike-gated run return a dictionary shaped like the SEP product boundary. This keeps downstream drawing and future benchmark code simple.

Key	Type	Meaning
raw_segmentation	2D int array	Relabeled MTOBJECTS segmentation before comparison-layer filtering.
filtered_segmentation	2D int array	Segmentation after geometry and size filters.
mask	2D bool array	Dilated foreground mask used for visual/profile cleaning.
cleaned	2D float array	Original image with mask pixels replaced by the finite unmasked median.
residual	2D float array	Original minus smoothed model, used for display and measurements.
rows	list[dict]	Per-label measurements and keep/reject state.
background_level / background_rms	float	MTOBJECTS estimated or supplied background values for audit display.

Comparison Readiness

- Detector-specific work is constrained to mtobjects_products and spike_gated_mtobjects_products.
- Rows contain detector-neutral fields: label, area, centroid, elongation, peak residual significance, centre distance, and kept.
- The output folder and PNG naming include MTOBJECTS-specific alpha and move_factor values.
- The display uses the same bar-aligned profiles and isophote helpers as the existing foreground-mask testers.

Error Handling and Setup

- If MTOBJECTS cannot import or load its shared libraries, the adapter raises a setup-focused error message.
- Non-finite pixels are replaced only for MTOBJECTS preprocessing, then restored to background in the segmentation.
- The max-tree C memory is released in a finally block after filtering.
- The script syntax-checks without MTOBJECTS installed because imports occur inside the runtime adapter.

Maintenance Notes

- For a formal SEP versus MTOBJECTS benchmark, call detector product functions directly and compare mask fraction, object counts, profile residuals, runtime, and manual review flags.
- Keep new detector adapters returning the same product contract rather than branching the plotting code.
- If MTOBJECTS is packaged differently later, update `mtobjects_context` only; the rest of the tester should not need to change.
- If replacement semantics change from median fill to interpolation/inpainting, implement that after product creation so detectors remain comparable.